

A Multiobjective Decision Support System for Multiresource Forest Management

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Abstract

A multiobjective and/or multiperson decision support system for analyzing multiresource forest management problems is developed in this paper. The procedure includes formulating the problem in a multiobjective and group decision making framework, and solving it using two solution techniques which consist of a distance-based compromise programming (CP) and a cooperative game theoretic approach of the Nash equilibrium type. The problem consists of five forest resources management objective functions to be maximized. Solving the problem using the two solution techniques enables determining a satisfactory compromise solution of the five forest resource management objectives. Sensitivity analysis of the two techniques shows compromise programming to be more sensitive to changes in the weight and the p-parameter of the technique while the cooperative game theoretic approach is relatively robust with respect to changes in the worst utility set.

Key words: compromise programming, cooperative game theory, decision support system, forest resources management, multiobjective analysis

Introduction

Forest ecosystems are composed of a complex mosaic of biophysical structures of diverse amenity and commodity resources such as timber, water, forage, wildlife, wilderness, soil (land), minerals, and other items of social, economic, cultural, and historical values intimately intertwined. The challenge in the management of such a system is to consider these values as interacting and in many cases interdependent components of the system and evaluate their individual responses and group interactions in a multiobjective management framework. Because of the interactions and interdependence, a change in one resource condition may have profound and often times conflicting influence on the status of the other resources in the forest system. Recognition of this kind of relationship among the different resources elements in a forest ecosystem is not new. Attempting to develop a reliable procedure for evaluating conflicting interaction among the different resources, however, is fairly new.

In an effort to minimize the adverse effects of manipulating one or more forest resources upon the remaining ones in a forest ecosystem and to promote the development of appropriate multiresource forest management schemes, a number of laws, regulations and Executive orders have been promulgated or issued by the United States Congress and the President.

These include the Multiple-Use Sustained Yield (MUSY) Act of 1960 which promotes the management of federal lands for multiple use and sustained yield of goods and services; the National Environmental Policy Act (NEPA) of 1969, which dictates the need for evaluating all proposed management activities for any potential environmental impact they may have; the Forest and Rangeland Renewable Resource Planning Act (RPA) of 1974 which requires assessment of current and future resources production capability of all national forests and rangelands and development of national programs to accomplish it, the National Forest Management Act (NFMA) of 1976 and many forest-related specific Executive Orders. The NFMA was designed to incorporate many of the important elements in the above laws, and stipulates the management of the national forests to provide multiple uses and products while at the same time accounting for any environmental impact of management as required by the NEPA 1969. Likewise, specific Executive Orders such as 11988 and 11990 have been issued to evaluate flood hazards of and protect wetlands from new projects on federal lands, respectively.

In spite of the wide recognition of the need for multipurpose and/or multiresource management of forest resources and the passage of several laws and issuing of various regulations to promote and practice it, however, there are only few signs to indicate the development of the needed technology to evaluate the most preferred mix. The purpose of this paper is to provide and demonstrate a procedure for interactive, multiobjective analysis of a forest ecosystem usable for group decision-making.

To demonstrate the procedure, this paper uses five resources response functions representing on-site wildlife habitat, forage yield, aesthetic value, on-site merchantable timber growth and water yield in the ponderosa pine forests of north-central Arizona. Availability of information and commensurability of the different decision variables are the main criteria for using only these resource elements in this paper. It is recognized that there may be many more resources and supporting elements such as treatment costs, water quality, flood hazard, recreational use, cultural preservation, and other economic and environmental implications of management that should be included in a multiobjective analysis of the forest resource management problems. However, we believe that about five decision-makers would represent most real group situations, as each has a specific subset of interests.

To present a clear picture of the analytical procedure used, the paper is organized as follows. A brief statement on the multiobjective formulation of the problem is presented, followed by a description of the techniques used. The particular multiobjective analytical techniques used are the distance-based compromise programming (Zeleny 1973, 1974, 1982; Tecle et al. 1988; Tecle 1992) and the Nash-Harsanyi type cooperative game theoretic approach (Nash 1950, 1953; Harsanyi 1963, 1977). An interactive decision support algorithm using these techniques is developed for evaluating trade-offs among the resources considered. The solution to the multiobjective problem using the two solution techniques is then obtained and sensitivity analyses with respect to weights and to different sets of worst utility values are performed for the compromise programming and the cooperative game theoretic approach, respectively. Finally, a brief summary and conclusions are provided. It is hoped that the procedures presented herewith will provide forest resources managers with a suitable tool for multiobjective evaluation of forest resources interactions during the planning and operational stages of management.

Problem formulation

Objective functions and decision variables

In this paper, the multiobjective problem consists of five time-invariant continuous objective functions which are constructed with the aim of maximizing the desired output for each resource element. Conceptually, this is meant to represent the desires and/or aspirations of different interest groups or decision makers. The decision makers may be federal agencies such as the Forest Service, the Bureau of Land Management, the Fish and Wildlife Service, or a state, county or native American government, or land owners or resource users such as lumber companies or ranchers. In many cases, many of these entities, environmental groups and other resources users may be conflicting agents in the multiobjective forest resources management decision making process.

In general, all the objective functions of the problem are concerned with determining an appropriate forest management scheme to satisfy simultaneously different societal needs of the forest system. The candidate management options consist of vegetation treatment levels ranging from zero to a complete harvest of the forest system. The objectives to be satisfied, on the other hand, address some of the major issues for which the forest system is being managed. The five particular objectives considered in this problem are individually described and formulated below. The decision variable in all of these objectives is tree basal area in square feet per acre. The decision makers are thus able to select the most satisfying tree density level for each objective function and the composite solution. The five objective functions and their mathematical formulation are described below.

Wildlife habitat condition. For the purpose of this paper, the response function for Abert squirrel habitat condition is used as an index to describe a general wildlife habitat. Such representation is clearly a gross oversimplification of the hundreds of different species of wildlife present in the ponderosa pine forest ecosystem of north central Arizona. In the absence of a better and more generalized index, however, selecting a single wildlife habitat that overlaps many other ones seems to be appropriate. Abert squirrel has such a habitat, and it is the most studied specific wildlife habitat in the southwestern ponderosa pine forest ecosystem (Ratcliff et al. 1975; Patton 1977; Ffolliott and Patton 1978; Patton et al. 1985; Dr. David Patton, personal communications).

The response function for Abert squirrel habitat condition in uneven-aged ponderosa pine forest ecosystem is developed on the basis of information obtained from Patton (1984) and McTague (1991). The wildlife habitat objective values are expressed qualitatively in ordinal units ranging from 1 to 5, in which 1 represents poor habitat quality and 5 stands for the best possible habitat quality condition associated with the corresponding basal areas. On this basis, the following equation, which is also graphically shown in Figure 1, has been constructed to represent the wildlife habitat response function or, alternatively, the utility of a wildlife interest group:

$$Z_1 = 0.8565 + 0.00623*X + 6.95e - 5*X^2 \quad [1]$$

where, Z_1 = ordinal wildlife (Abert squirrel) habitat level;
 X = tree basal area in sq.ft./acre.

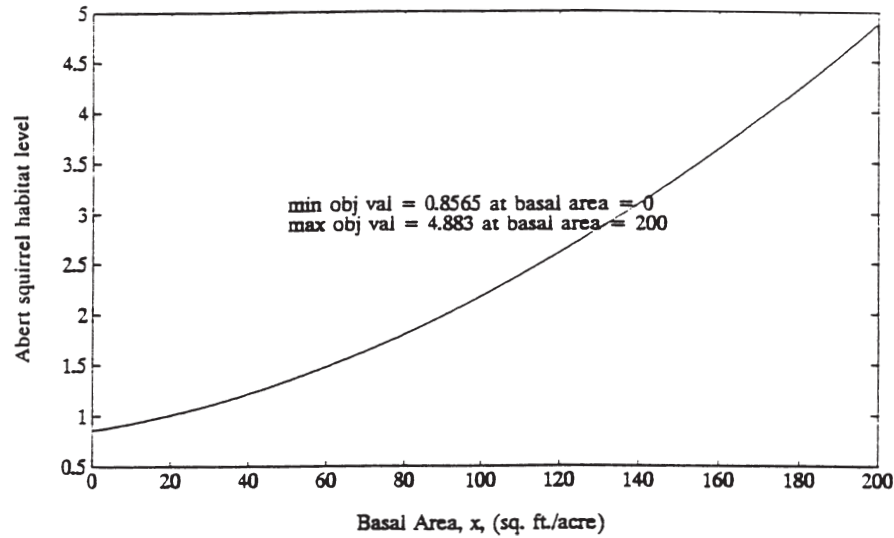


Figure 1. Wildlife (Abert squirrel) habitat condition versus tree basal area (in sq.ft./acre).

Herbage production. Many forest density/herbage production relationships applicable to southwestern forest ecosystems are known in the literature (Brown et al. 1974; Ffolliott and Clary 1974, 1982; Patton 1974; Mitchell and Pickens 1985). In this paper, the ECOSIM model (Rogers et al. 1984) for herbage production function developed for southwestern ponderosa pine type and described in Covington and Fox (1991) is used. The equation developed for the area using a large data set by Solomon (1985) is written as

$$Z_2 = 45 + (24*Prec + 55*Dep) (\exp (-0.0289*X)) \quad [2]$$

where, Z_2 = herbage production in lb/acre;
 $Prec$ = annual precipitation in inches/year;
 Dep = depth of soil to impedance layer (in inches);
 X = tree basal area in sq.ft./acre.

Note that the values for $Prec$. and Dep . have to be pre-decided. Figure 2 shows the response function f or herbage production used in this study.

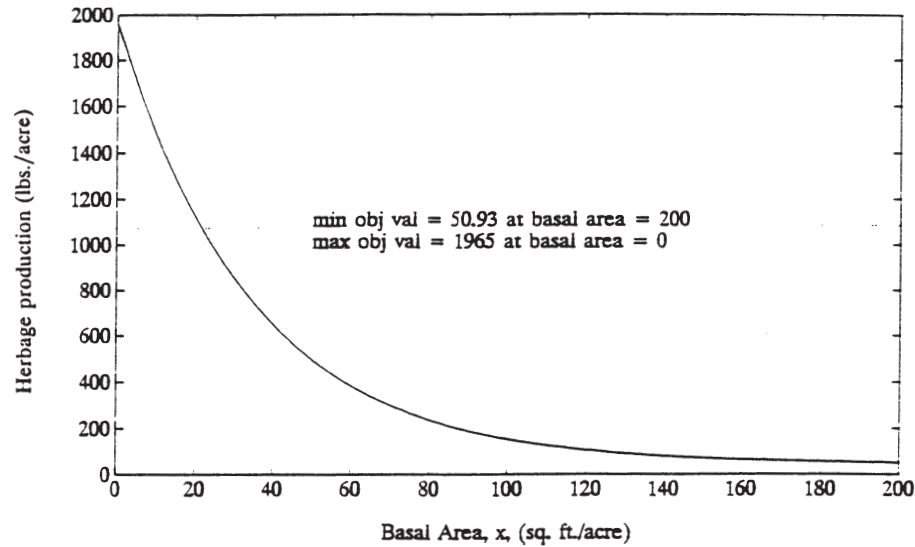


Figure 2. Herbage production response to changes in residual tree basal area (in sq.ft./acre).

Aesthetic value. Substantial modeling efforts to predict the aesthetic quality implications of a ponderosa pine forest management have been made recently (Brown et al. 1974; Brown and Daniel 1984, 1986). Multivariate models are provided to express the aesthetic values of different levels of timber stands in the latter two studies while a univariate response function is provided in the first. A modification of the response function for aesthetic quality developed in Brown et al. (1974) is used in this paper. As in the wildlife habitat condition, the aesthetic values are expressed in ordinal, qualitative levels, namely scenic beauty index (SBE). These levels represent the subjective feelings of individual visitors to the forest system. Such feelings are regressed against residual forest tree density in basal area to produce the equation stated below. A graphical representation of this equation is shown in Figure 3.

$$Z_3 = 1.144 - 0.001974 * X + 0.0004718 * X^2 - 4.356e-6 * X^3 + 1.059e-8 * X^4 \quad [3]$$

Where, Z_3 = Recreation values (in scenic beauty index);
 X = tree basal area in sq.ft./acre.

On-site merchantable timber volume. The response function for on-site timber volume is developed using data from the ponderosa pine forest area of northern Arizona presented in Ronco et al. (1985). The response function in this objective is expressed in terms of on-site merchantable timber growth volume (in cu.ft./acre). The fitted response function is graphically shown in Figure 4.

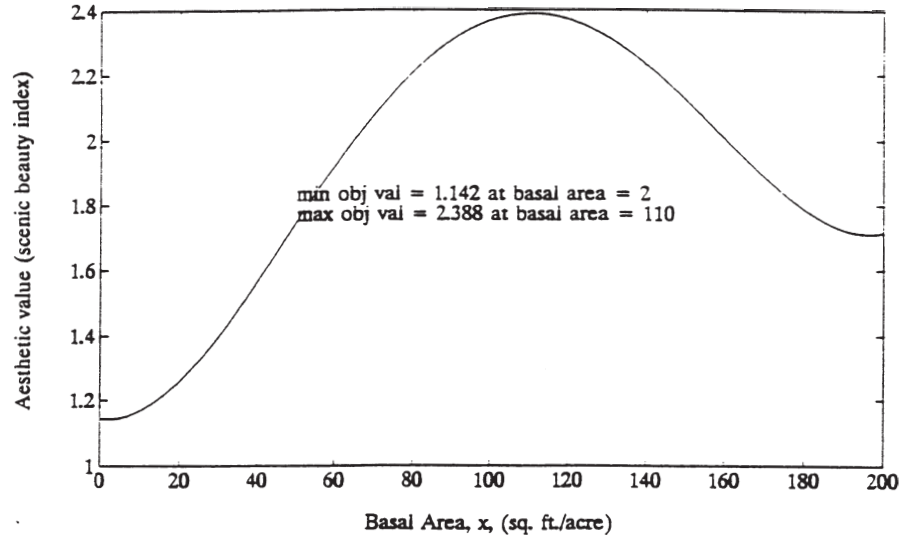


Figure 3. Forest aesthetic value response to changes in residual tree basal area (in sq.ft./acre).

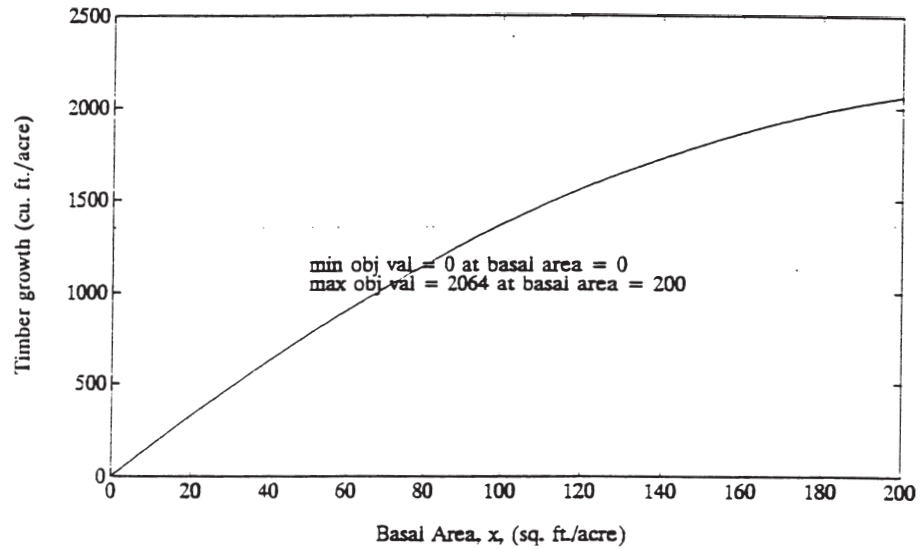


Figure 4. On-site merchantable timber growth response to changes in residual tree basal area (in sq.ft./acre).

$$Z_4 = 17.06 \cdot X - 0.0336 \cdot X^2$$

Where, Z_4 = merchantable timber growth volume (in cu.ft./acre);
 X = tree basal area in sq.ft./acre.

Water yield. The water yield response function used in this paper is the modified version of the Baker-Kovner model described in Rogers et al. (1984) and Tecle (1991). It is the model for estimating water yield response to harvesting currently used by the Forest Service in its southwestern ponderosa pine forest management planning. The equation is written as

$$Z_5 = 1.19 \{ -5.72 + 0.83 * P_w + 0.42 * r - 0.24 * r * P_w^{0.92} - 0.007 * P_w^2 * [1 - \exp(-X/45)]^3 \} - 0.47 \quad [5]$$

where, Z_5 = annual streamflow in inches;
 P_w = winter (1 Oct.–30 April) precipitation in inches;
 r = insolation index;
 X = tree basal area in sq.ft./acre.

Figure 5 sketches this modified Baker-Kovner response function for water yield. Note that the r and P_w values in this response function are statistical long term averages. This was done in order to have a two-dimensional response function of water yield versus tree density in basal area.

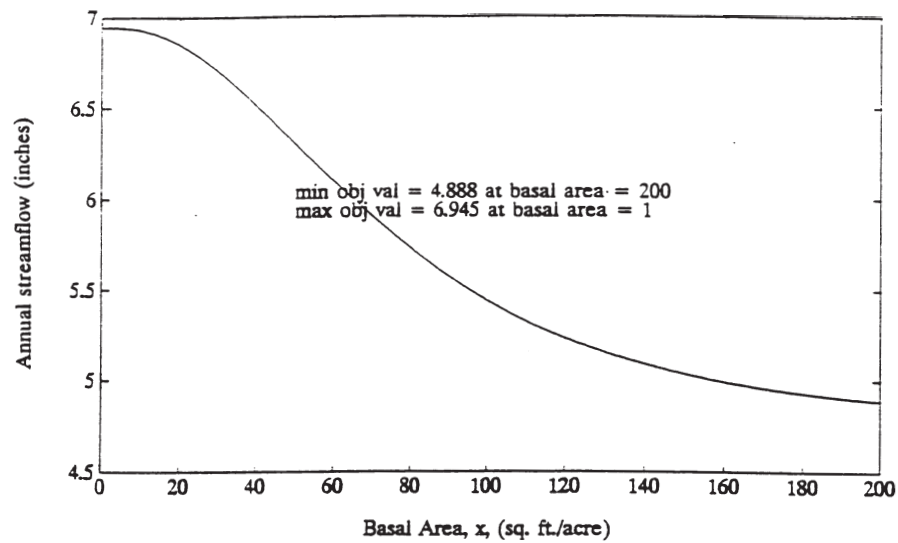


Figure 5. Water yield response to changes in residual tree basal area (in sq.ft./acre).

Since more of the resources represented by these response functions is usually felt to be desirable by different decision makers, they all are to be maximized during the evaluation process. A combined graphical representation of the standardized forms of all five objective functions quantitatively described in equations 1 through 5 is presented in Figure 6 to provide the reader with a visual comparison of the shapes of these functions. The solution techniques used to evaluate simultaneously these objectives are described next.

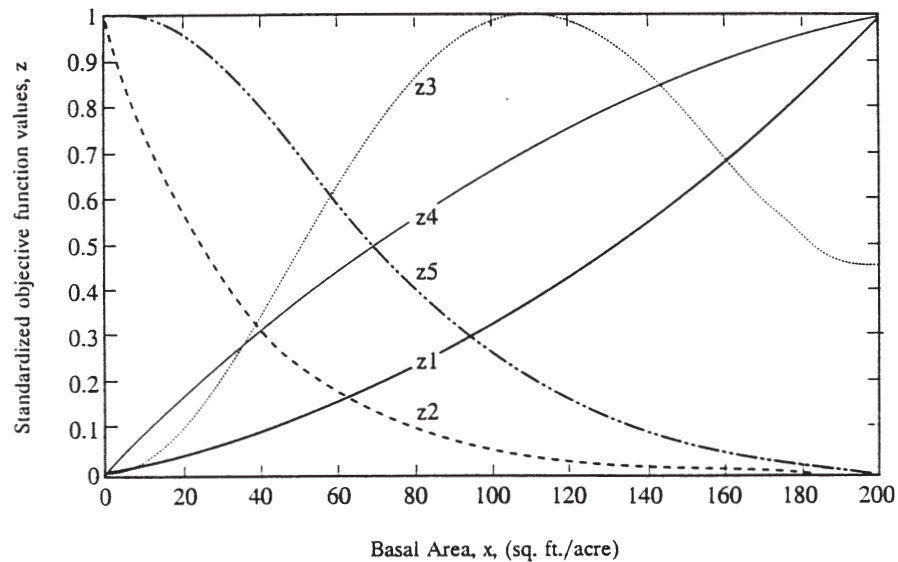


Figure 6. Graphical comparison of the five resources response functions. [Resource yield versus residual tree basal area (in sq.ft./acre).]

Techniques

The concept of nondominance is used to determine a solution to the problem under consideration. A nondominated solution can be defined as a feasible solution to a multiobjective problem for which there exists no other feasible solution that will cause improvement in any one of the objectives without making at least one other objective worse (Goicoechea et al. 1982; Szidarovszky et al. 1986; Tecle et al. 1988; Tecle and Duckstein 1993).

Specifically, two different multiobjective solution techniques which are based on this concept of nondominance are used to evaluate the problem. These are a distance-based technique, compromise programming and a cooperative game theoretic approach of the Nash equilibrium type. On the basis of the authors' experience, both techniques are considered suitable for evaluating the forest resources management problem considered (Szidarovszky et al. 1984; Tecle et al. 1988; Tecle 1992). There is also some gain in popularity in the use of MC group decision making procedures in general, and these two techniques in particular, in the literature as indicated recently in Hipel (1992), Hobbs et al. (1992), Iz (1992), Tecle (1992), and Iz and Gardiner (1993). The particular details of the two techniques are briefly described below.

Compromise programming. Compromise programming is designed to identify the particular solutions from among the Pareto optimal (nondominated) set which are the closest, by some distance measure, to a generally infeasible ideal solution (or utopian point). In this regard, obtaining compromise solutions can be viewed as minimizing a decision maker's regret of not obtaining an ideal solution.

The technique allows the use of a two-fold weighting scheme in the evaluation process. One set of weights are introduced to express the decision maker's preference structure among the different objectives under consideration, while the other allows one to indirectly attach relative weights to deviations from a target (or ideal solution) value.

The measure of closeness is defined by a family of distance metrics, called the weighted l_p metrics. The general formulation of this technique can be expressed as

$$\min \{I_p = [\sum_{i=1}^I W_i^p (Z_i^* - Z_{ij})^p]^{1/p}, j = 1, \dots, J \} \quad [6]$$

here Z_{ij} is the normalized value of objective i for treatment j , and Z_i^* is the best value for objective i ; I is the number of objectives and J is the total number of treatments considered. In this procedure the set of maximum values of all objective functions constitute the ideal solution to the multiobjective problem.

Normalization of the objective functions is generally needed to make all the objective function values commensurable and thereby avoid scale effect in the compromise solution. Using Z_i^{**} as the worst value for objective i , the following expression is used to normalize the objective functions:

$$Z_{ij} = \frac{Z_{ij} - Z_i^{**}}{Z_i^* - Z_i^{**}}, i = 1, \dots, I \text{ and } j = 1, \dots, J \quad [7]$$

which always guarantees values of Z_{ij} within the $[0,1]$ range.

The weight w_i in equation [6] signifies the importance of objective i in comparison to the other objectives and p is the metric parameter. Varying the parameter p from 1 to infinity, allows one to move from minimizing the sum of individual regrets (i.e. having a perfect compensation among the objectives) to minimizing the maximum regret (i.e. having a no compensation among the objectives) in the decision making process. The choice of a particular value of this compensation parameter p depends on the type of problem and desired solution. In general, the greater the conflict between players is, the smaller the possible compensation becomes. Detailed discussion on the effect of this parameter on the compromise solution to a multiobjective problem may be found in Zeleny (1974, 1982), Goicoechea et al. (1982), and Szidarovszky et al. (1986).

Cooperative game theoretic approach. To use the cooperative game theoretic approach, the different objectives under evaluation are considered to represent competing interests of different decision makers (or players). Furthermore, use of the procedure to obtain a multiobjective solution requires one to assume some kind of cooperation or interaction among the different players (Teclé et al. 1988).

A particular procedure that has been useful for dealing with n-person cooperative game type of problems is the Nash-Harsanyi Bargaining Model (Nash 1950, 1953; Harsanyi 1963, 1977). Application of this model requires development of utility functions for each objective under consideration. In this problem, the normalized objective function values are used as the utility functions since the values lie within the $[0,1]$ range. Note

that since every player is expected to maximize his/her own utility, objectives that have to be maximized can directly be used as the desired utility functions while those objectives that have to be minimized are converted to utility functions by taking their negative values.

The Nash-Harsanyi model can be expressed in the following form:

$$Z_i^* = \max_{j=1}^J \pi (Z_{ij} - S_i), i = 1, \dots, I \quad [8]$$

$$\text{s.t. } \begin{aligned} Z_{ij} &\geq s_j, i = 1, \dots, I \text{ and } j = 1, \dots, J \\ Z_i^*, Z_i &\in P, \text{ and } s \in P \end{aligned} \quad [9]$$

in which Z_i ($Z_i = \{Z_{ij}, \dots, Z_{ij}\}$) represents the payoff vector (in terms of utility values), Z_i^* is the most satisfying level for each objective and s ($s_1, \dots, s_{1j}$) is the disagreement payoff vector (i.e., the worst payoff vector) in the particular payoff space P under consideration. The disagreement payoff vector represents the payoffs individual players would receive if they cannot reach any agreement on the final strategies, provided there exists at least one feasible solution. In this procedure the disagreement vectors are pre-established, and the resulting solution/agreement is assumed to be unanimous.

There are some requirements which must be met in using the Nash-Harsanyi model, namely Nash's axioms should be satisfied: feasibility, collective rationality, joint efficiency, symmetry, linear invariance and independence of irrelevant alternatives. The detailed description of these axioms can be found in Goicoechea et al. (1982) and Tecle et al. (1989). The solutions obtained using the cooperative game theoretic approach and those using the compromise programming technique are discussed below.

Analysis of results and discussion

The solutions in this study are determined using interactive computer algorithms written for both compromise programming and the Nash-Harsanyi cooperative game theoretic approach. The algorithms are built as two components of a single interactive decision support system to provide natural resources management decision makers with two different solution approaches to a multiobjective problem. The compromise programming solutions are determined on the basis of their closeness to an ideal or the best possible solution while the cooperative game solutions represent the furthest unanimously acceptable solution from the disagreement vector point.

Solutions. The results of analyzing the problem using the compromise programming algorithm are shown graphically in Figures 7a, 7b and 7c. These figures are the compromise solutions of the problem for compensation factor, p values of 1, 2 and ∞ , respectively. These solutions were determined with all five objectives having equal weights. Under this condition, the solutions were determined to be the maximum basal area (here 200 sq.ft./acre) for the perfect compensation (i.e. $p = 1$), 78 sq.ft./acre for little compensation (i.e. $p = 2$) and 62 sq.ft./acre for the no compensation (i.e. $p = \infty$)

In the case of perfect compensation ($p = 1$), we see that the values of objectives 1 (wildlife), 3 (recreation) and 4 (timber yield) fully compensate the values of objectives 2 (range) and 5 (water yield) resulting in the optimal basal area of 200 sq.ft./acre. When only little compensation among the objectives is allowed, however, then the resulting optimal basal area of 78 sq.ft./acre seems to be more reasonable. Similarly, when zero compensation is allowed, corresponding to minimizing the maximum regrets of decision makers, the solution becomes a compromise basal area level of 62 sq.ft./acre. Under these circumstances, one can conclude that for p values ranging from 2 to ∞ , the solutions lie in between 78 to 62 sq.ft./acre. From the experience of the authors this value is reasonable for real world problems such as the one under consideration.

A set of solutions obtained using the Nash-Harsanyi cooperative game theoretic approach are shown in Figures 8a, 8b, 8c and 8d. This set consists of the cooperative game solutions for **0**, **0.05**, **0.10** and **0.15**. The corresponding optimal basal areas are 70, 66, 63 and 62 sq.ft./acre. Note that, in this case, each vector, for example, **0**, represents the worst utility values $\{0,0,0,0,0\}$. Also, note that the feasible set decreases as worst utility vector increases. This means that there is not much space left for negotiation as the players' minimum acceptable objective function level increases.

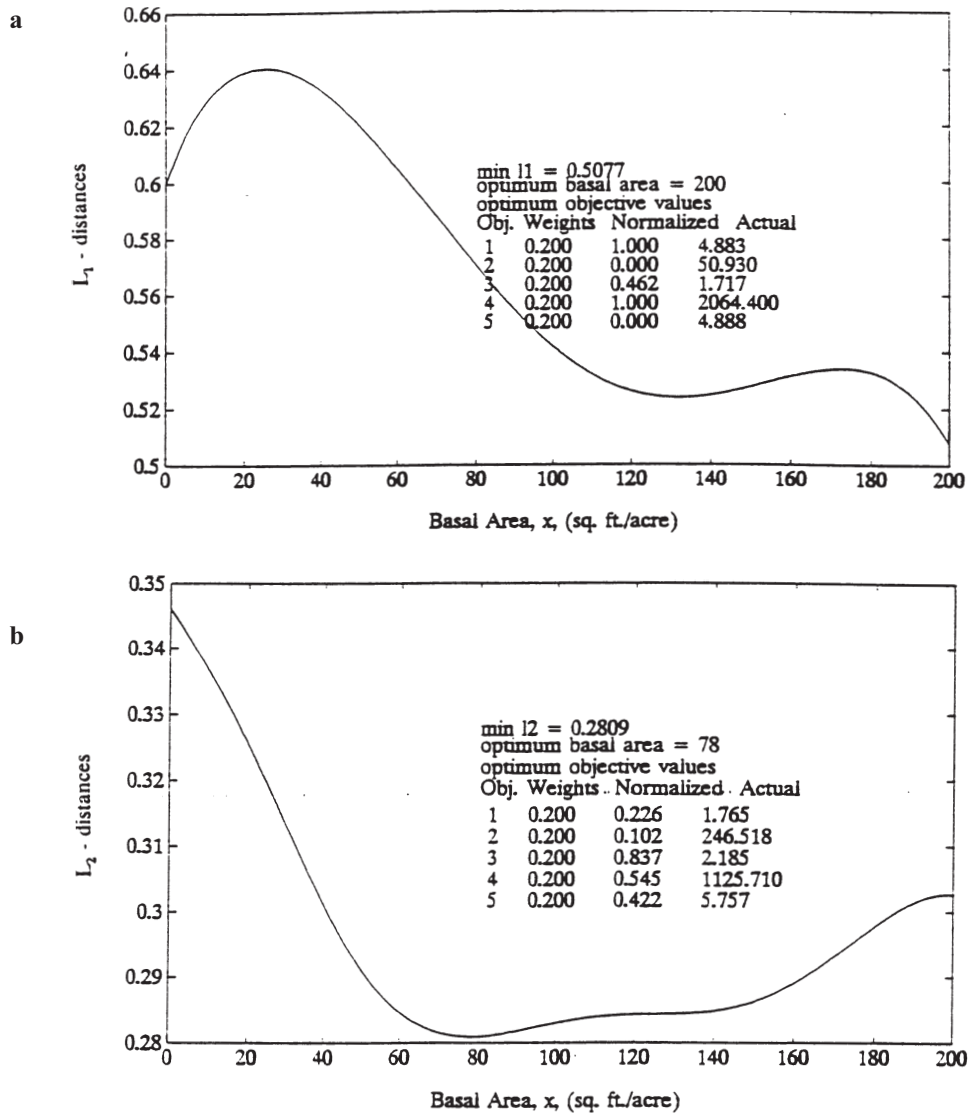
Depending on the structural relationship of the objective or utility functions used, there is a limit to the level of maximum worst utility vector players can accept. In the present case study, up to 0.16 of worst utility vector values can give at least one feasible solution. However, one can trade off one objective's worst utility value with another to allow assignment of different utility values to the objectives. When equal worst utility values are used, the entire cooperative game solutions become tree basal area levels in the range of 62 to 70 sq.ft./acre.

Note that when the decision makers want to achieve the maximum utility values for all the objectives, then the cooperative game solution and compromise solution for no compensation match. Comparing the optimal basal areas obtained using cooperative game theoretic approach and compromise programming using p values between 2 and ∞ shows that basal area levels ranging from 62 to 78 sq.ft./acre are satisfactory to all the decision makers when every objective is weighted equally. Clearly a negotiation is necessary to reach such a compromise.

Sensitivity analysis

Sensitivity analyses were performed to test the robustness of each algorithm with respect to changes in its respective parameter values. The solutions obtained using compromise programming for different combinations of weights is given in Table 1. Likewise, the optimal basal area for the cooperative game theoretic approach under different combinations of worst utility values of the objectives are shown in Table 2.

The tree basal area values in Table 1 indicate that the sensitivity of the distance-based technique compromise programming to changes in weights varies from very high to moderate with an increase in the value of the parameter p . As shown in the second row of Table 1, weighting objectives herbage production and water yield which are favorable



under lower basal area a little more than the other objectives changes the value of the resulting basal area drastically (almost to a minimum). When wildlife habitat and onsite timber volume which are favorable under denser forest conditions are weighed higher, however, the preferred tree basal area becomes much higher (see row 3 in Table 1). Table 2, on the other hand, shows that the cooperative game theoretical approach is not very sensitive to changes in worst utility values. One of the reasons may be that in this approach, one really cannot choose very high worst utility values since such a choice may result in an infeasible solution set.

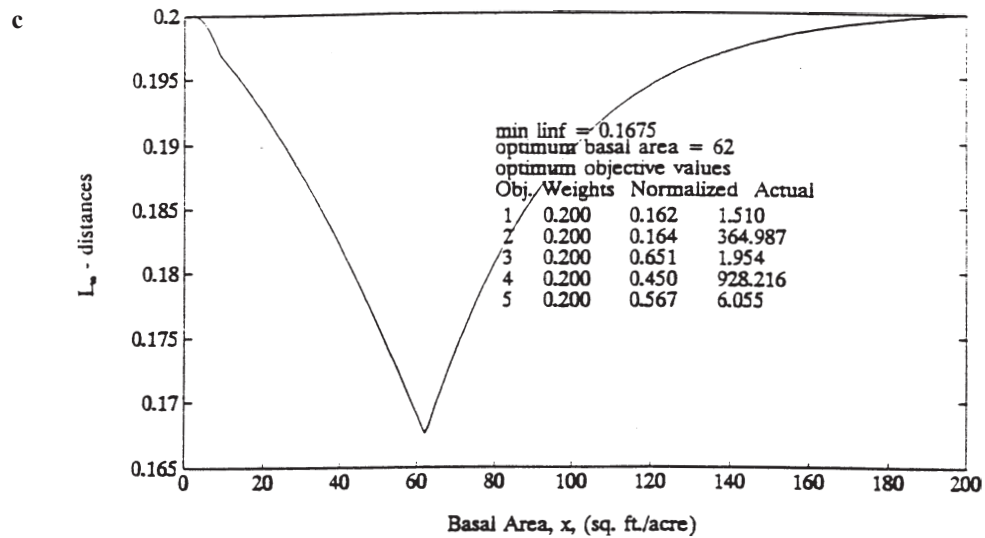


Figure 7a-c. Compromise programming solutions for (a) perfect compensation ($p = 1$), (b) little compensation ($p = 2$), and (c) no compensation, versus residual tree basal area (in sq.ft./acre).

Table 1. Sensitivity analysis of the compromise programming technique

Row	Weights					Optimal basal area (sq.ft./acre)		
	Objectives					Metric parameter		
	1	2	3	4	5	$p = 1$	$p = 2$	$p = \text{inf.}$
1	10	10	10	10	10	200	78	62
2	20	30	20	10	40	0	2	33
3	30	20	30	40	10	200	156	107
4	30	30	20	50	40	200	78	68
5	50	40	60	30	30	126	143	87

Table 2. Sensitivity analysis of the cooperative game theoretic procedure to varying worst utility values

Row	Worst set of utilities					Optimal basal area (sq.ft./acre)
	Objectives					
	1	2	3	4	5	
1	0	0	0	0	0	70
2	5	5	5	5	5	66
3	10	10	10	10	10	63
4	15	15	15	15	15	62
5	20	10	15	25	10	75
6	10	20	15	10	25	50
7	10	5	25	5	10	72
8	10	25	25	25	10	46

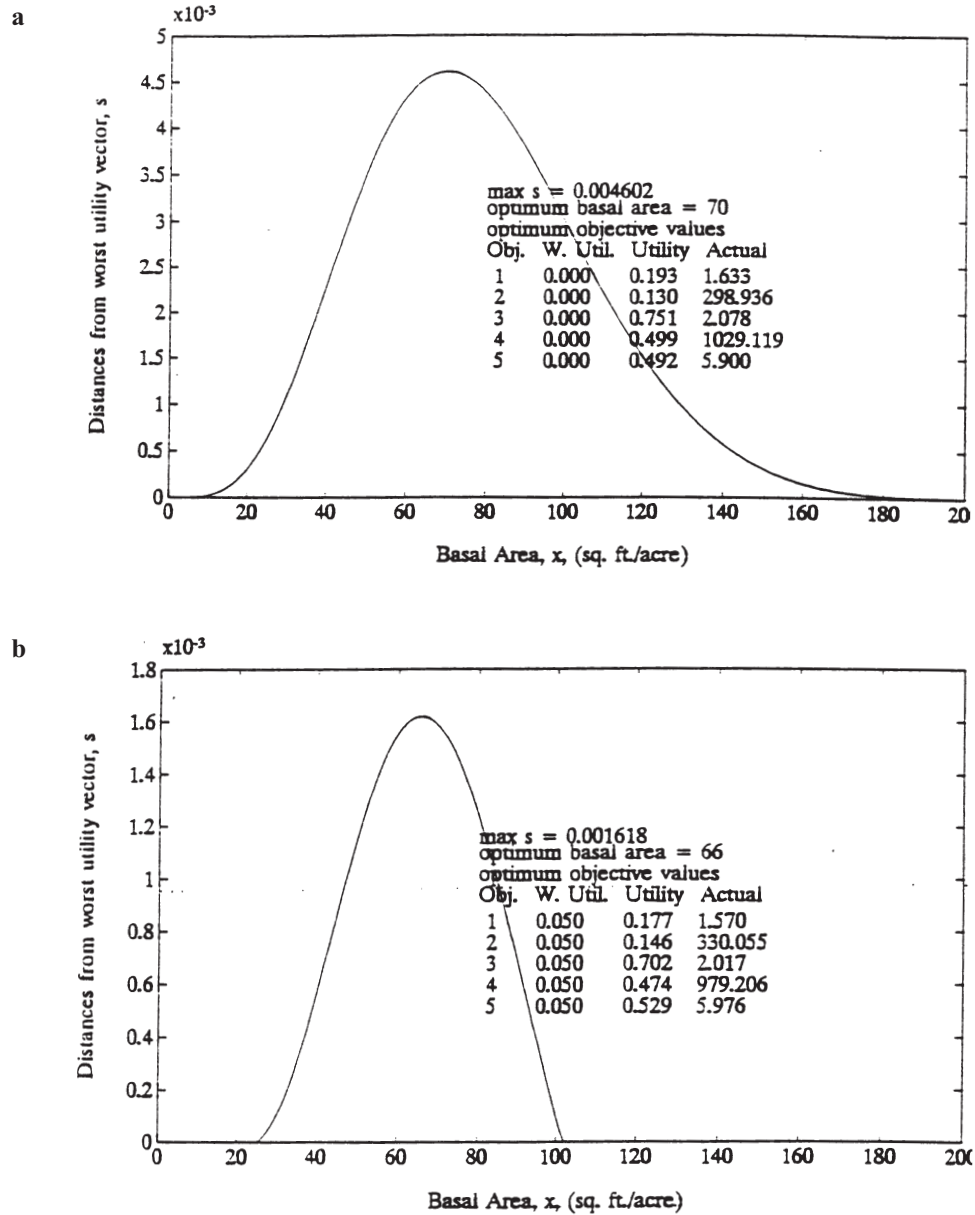


Figure 8a and b. Cooperative game theoretic solution of the multiobjective problem using worst utility vector values of (a) 0 and (b) 0.05.

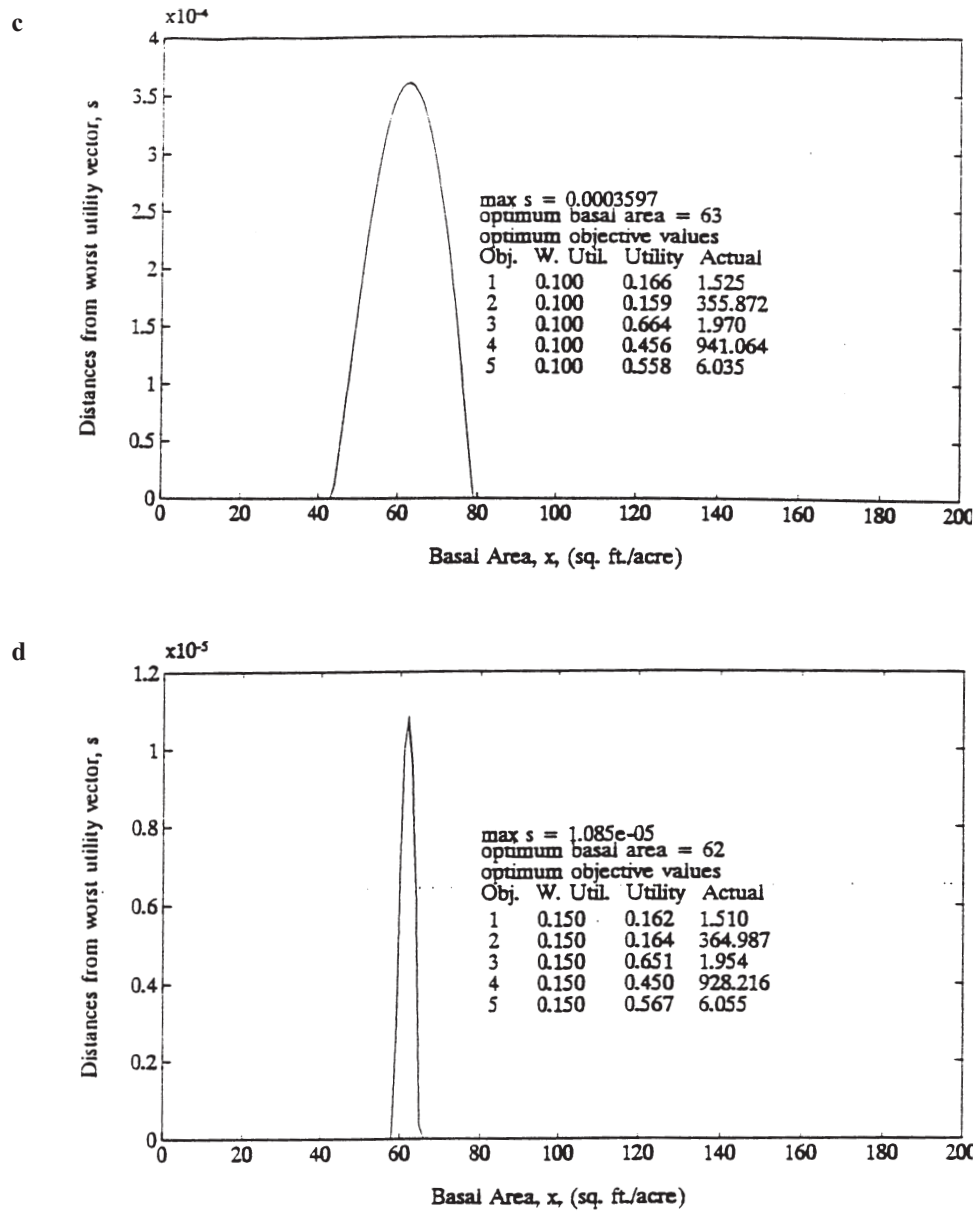


Figure 8c and d. Cooperative game theoretic solution of the multiobjective problem using worst utility vector values of (a) 0.10 and (b) 0.15.

Under the conditions stated in this paper, it seems that game theoretic approach is more robust than compromise programming. This, however, may not be a general statement.

Summary and conclusions

An interactive multiobjective decision support system has been developed to aid in a multiresource forest management decision making process. The decision support system combines algorithmic use of the two different multiobjective decision making aids of compromise programming and the Nash-Harsanyi cooperative game theoretic approach. To demonstrate its utility, the software has been used to solve a particular real world case study problem.

The case study consists of a multiobjective ponderosa pine forest resources management problem in north-central Arizona. Even though the forest system is managed to promote the development and best use of many more individual resources and control of environmentally undesirable situations such as sediment yield and flood hazard, for example, this paper discusses management of the forest system to achieve only five resources improvement objectives: wildlife habitat, herbage production, aesthetic value, merchantable timber volume and water yield.

The level of manipulation of the forest system to achieve these objectives is evaluated using the two multiobjective decision making aids. The analyses results show that the optimal basal area for satisfying, in a compromise sense, the above resources objectives lies in the range of 62 to 78 sq.ft./acre in a 200 sq.ft./acre basal area forest system. It must be remembered, however, that the best management schemes can be obtained by including all the relevant forest resources management objectives in the analysis. Such objectives should include minimization of undesirable outcomes of management like cost, water quality degradation, flood hazard and air quality, or maximization of desirable outcomes like preservation of cultural artifacts, wilderness condition, visitor safety, mineral extraction and wildlife diversity. Since the main objective in this paper is to demonstrate the development and use of a multiobjective analytical procedure for multiresource forest system management, the focus should be on the algorithmic soundness of the procedure to serve the purpose. In this regard, the software as designed seems to be a useful and reliable decision support system.

The software, FORMDSS, is designed to provide natural resources management decision-makers with an interactive, flexible and easy to handle decision support system (Shrestha et al. 1992). This means it is a system that readily permits changing the number and form of the objective functions as well as the relative form of the decision makers' preference structure. It has the capacity to handle as many objectives as desired and can solve discrete or continuous problems. It has also a graphical capability to visually show both the structure of the objective functions individually as in figures 1 through 5 and as a group (Figure 6), as well as the form of the final compromise or acceptable solutions. And

if in the latter case, the decision makers are not satisfied with what they see, then they can interactively either provide different preference structures in the compromise programming algorithm or change their worst utility vector in the case of the cooperative game approach. In any case, since the procedure allows free interaction among the decision-makers during the evaluation process, the derived solutions are expected to be satisfying to all those involved.

Acknowledgements

This study has been partially supported by funding from the office of Graduate Studies and Research under the Organized Research Program, Northern Arizona University, Flagstaff, Arizona, and by a grant from the USDA/CSRS McIntire-Stennis Program (ARZZ-NAU-32).

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